Assignment 04 Report

Richard Neureuther 01/1001286

Introduction

In this assignment, we are tasked with fine-tuning a Large Language Model (LLM) to perform classification on speeches from the Austrian parliament and comparing its performance to the default, uninformed model. Austria has a pluralist party system with several parties representing different social cleavages and interests. Party orientations range from center-left or left, such as '*JETZT'* and '*Grüne'*, to right-wing and right-wing populist, as in the case of the '*FPÖ'*, making both tasks multi-class classification tasks with six and five classes respectively. The assignment is divided into two tasks: first, parliamentary speeches are classified according to the political party of the speaker; second, according to the political orientation of the party. The small pre-trained generative language model Qwen 3 0.6B is used to classify speeches first without any fine-tuning, and then after being fine-tuned using instruction fine-tuning as well as LoRA (Low-Rank Adaptation).

Data Analysis

The ParlaMint parliamentary corpus dataset is used, which contains speeches from various European parliaments. Specifically from the Austrian parliament covering the years 2018 to 2022. Only speeches with more than 50 words were included, and those of a procedural or administrative nature were excluded. This was achieved by filtering the "Speaker_role" variable and retaining only regular speakers, while removing speeches delivered by the parliamentary president or guests. In addition, any speeches where the two target variables "Speaker_party" or "Party_orientation" were missing were also removed, as they were not suitable for the classification task. Finally, the dataset was checked for duplicates by examining both the "Text_ID" variable and the speech content. No duplicates were found.

After cleaning, a total of 16,396 speeches are available in the dataset. The number of speeches varies significantly by party. For example, "JETZT" has around 500 speeches due to the party's short time in parliament, while other parties range from approximately 2,000 (GRÜNE) to about 5,000 (ÖVP). Despite "JETZT" being a significant minority class, the overall class distribution remains within an acceptable range. The second target variable, "Party_orientation," also shows some skewness. The smallest class, "Mitte" (Center), has around 2,000 cases, while the majority class, "Mitte-rechts bis Rechts" (Center-Right to Right), includes about 5,000 cases. The second most common class is "Mitte-Links" (Center-Left), 4000 cases. This skew is less pronounced compared to the first target variable.

The speeches contain an average of 3,759 characters and 552 words, with approximately 295 unique words used per speech with some very long outliers in speeches of parties. The total vocabulary size across the entire corpus is 282,874. After removing common German stopwords using the Natural Language Toolkit (NLTK), along with some additional custom stopwords such as "abg" (short for Abgeordnete) and "Beifall," the five most common remaining words are: 'herr' (29,714), 'övp' (23,783), 'frau' (20,198), 'österreich' (19,769), and 'menschen' (18,911). All words were converted to lower case.

For model evaluation and fine-tuning, the dataset was split into a training set (70%), a validation set (15%), and a test set (15%), resulting in 11,477, 2,459, and 2,460 speeches, respectively. Fine-tuning the model on the full dataset exceeded Colab's GPU memory limits and triggered CUDA out-of-memory errors. To address this, the dataset was reduced to 30 percent to ensure the training process fit within the available resources. This approach also reduced the significant time required for evaluation and fine-tuning. To preserve the natural distribution of speeches as observed in a real parliamentary setting, stratified random sampling was used. For the first task, the sampling preserved the distribution by *Speaker_party*, and for the second task, by *Party_orientation*.

Methods

Qwen3, a small pre-trained generative language model with 0.6 billion parameters, was used for this task. For both tasks, a base Qwen3 model was first used to establish a performance baseline through a zero-shot classification approach. Thinking was disabled for all tasks. The prompt for the first task was:

"You are an expert political scientist. Classify the following speech from the Austrian Parliament into the political party of the speaker. The options are: 'ÖVP', 'SPÖ', 'FPÖ', 'NEOS', 'JETZT', 'GRÜNE'. Answer with only the party abbreviation.".

This was then followed by the first 750 characters of the respective speech. This character limit was chosen in the interest of keeping evaluation and fine-tuning times within a sensible time frame. The max length of input was set to 512 tokens which should be enough to accommodate both the prompt and the speech limited to 750 chars. The prompt for the second task was similar to the first:

"You are an expert [...]. Classify [...] the ideological orientation of the speaker. The options are: 'Links', 'Moderat-Links', 'Mitte', 'Moderat-Rechts', 'Rechts'. Answer with only one of the available classes."

The options were reiterated in the prompt, as the model initially struggled to produce only valid class names. As before, the prompt was followed by the first 750 characters of the speech. It is worth noting that the class names were simplified from the original labels while keeping the meaning intact because the model had difficulty handling the repeated use of "Mitte" and the longer labels for moderate-left and moderate-right categories. Evaluation on the base model was then performed on the respective validation sets. The fine-tuning pipeline was kept identical for both classification tasks. First, an instruction dataset was created by formatting speeches into instruction–response pairs, with each sample including the prompt, the corresponding speech, and the correct class label. In total, 3,441 speeches from the training dataset and 2,459 speeches from the validation set were formatted in this way. Next, Low-Rank Adaptation (LoRA) fine-tuning was performed using the *peft* library. The task was specified as causal language modelling, with inference mode disabled. Low-rank adapters were configured with a rank of 16, a scaling factor of 32, and a dropout rate of 0.1. This setup introduced a total of 10,092,544 trainable parameters, which were added to the frozen parameters of the base model, resulting in a total model size of 606,142,464 parameters.

Training was conducted for 3 epochs using a batch size of 2, again due to GPU memory limitations, with 16 gradient accumulation steps, resulting in an effective batch size of 32. The learning rate was set to 0.0001, with 20 warmup steps. The default AdamW optimizer form HuggingFace was used. Mixed precision training was applied, and the maximum gradient norm was set to 1 to prevent exploding gradients. Evaluation was performed on the validation set every 100 steps. The training and validation loss for both models can be found in Table 1 below.

Table 1: Loss Table	**Speaker Party**		**Party Orientation**	
	Training	Validation	Base	Finetuned
Step 100	1.885100	1.875422	1.706000	1.708686
Step 200	1.790900	1.812907	1.618700	1.651057
Step 300	1.723700	1.792790	1.588300	1.633707

Results

The base models struggled significantly with the classification task. Without domain knowledge and fine-tuning, the zero-shot classification task was not solvable, as evidenced by the performance metrics presented in Table 2. The base model labelled almost every speech as SPÖ in the initial task, resulting in a misleading accuracy of 0.2342 and a more informative F1 score of 0.09. This outcome suggests that the LLM either lacked sufficient contextual understanding to differentiate between political parties or was unable to extract the latent features necessary for accurate party classification. The same can be said for the second task, where the base model predicted "Moderat-Rechts" for every case. Again, accuracy is misleading, and the F1 score of 0.14 provides a clearer picture of performance. To save space, the confusion matrices for the two base models, which offer little additional insight, are not included in this report.

Table 2: Performance	**Speaker Party**		**Party Orientation**	
	Base	Fine-tuned	Base	Finetuned
Accuracy	0.2342	0.460	0.3016	0.255
F-1 Score *(weighted avg)*	**0.09**	**0.43**	**0.14**	**0.17**

The fine-tuned model showed significantly better performance on the first task but still leaves much to be desired, with an F1-score of 0.43. The confusion matrix for the model can be seen in Fig. 1. It performed moderately well in predicting the majority class, ÖVP, with some success in predicting FPÖ cases, with F1-scores of 0.59 and 0.42, respectively. Surprisingly, the performance on the second most common class, SPÖ, was worse than on FPÖ. This suggests that the model did learn some latent features about the speeches: while the performance on ÖVP could be partly attributed to its status as the majority class, the improved results on FPÖ indicate that the model captured some meaningful features associated with that class. The other classes were mostly misclassified as ÖVP, likely due to majority status, with NEOS being misclassified as FPÖ. This performance again demonstrates that the model lacked deep understanding of the domain, likely due to the limited fine-tuning time due to memory restrictions.

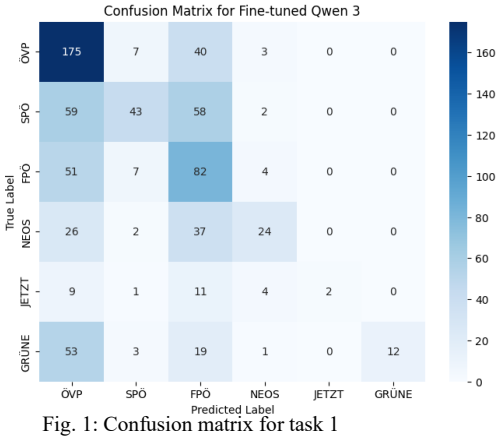

Fig. 1: Confusion matrix for task 1

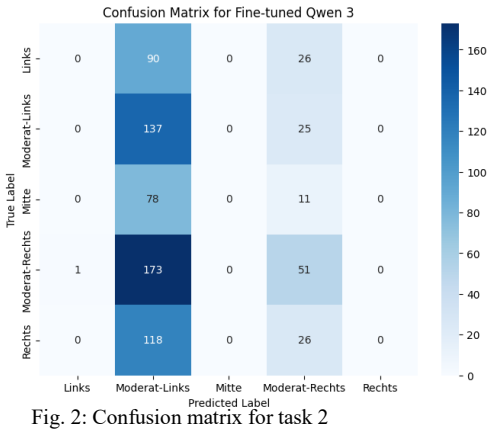

Fig. 2: Confusion matrix for task 2

For the second task, the classification behaviour did change; however, no significant improvement was observed, with an F1-score of 0.17. The model defaulted to classifying almost all cases as "Moderat-Links," with a few cases labelled as "Moderat-Rechts," the previous default class. This suggests the model struggled to identify phrases or terms that clearly distinguish between the various political orientations, capturing only some general tendencies. To improve performance, the task could potentially be reframed as a binary classification problem, distinguishing only between left and right. Since the model appeared to recognize at least some directional cues, this approach might help it first learn the broader rhetorical differences between the two orientations and then be further trained to capture more nuanced distinctions within each political direction.

Discussion and Conclusions

Potential applications include comparing political rhetoric across systems by analyzing emerging shared language among parties (e.g., increased populist rhetoric or focus on social issues). With the rise of new conservative-left movements (e.g., BSW in Germany), one could examine whether their rhetoric aligns with that of established parties from other ideological camps (e.g., CDU or AfD) on social and cultural issues. This may reveal trends of ideological blending and realignment. A key limitation is that patterns learned from one political system may not transfer well to others. For example, far-right rhetoric differs significantly between Germany and the United States. Additionally, subtle linguistic cues that indicate political orientation in one language may be absent or differently expressed in another, limiting model effectiveness. Such challenges exist even between closely related settings, like the German and Austrian parliaments, where similar ideas may be conveyed through different idioms. The main findings were discussed in the Results section. One key methodological insight is that prompting has a significant impact, particularly in the second task where the model struggled. Reiterating and simplifying valid labels improved output quality, even if predictions were still incorrect.